# Let the computer search: review

## Optimization · Level 3 · Unit check



# In this unit

- Try them all, or think ahead
- Shortest paths
- The bus route
- Timetables with CP-SAT

# The big ideas

- Brute force is sure but explodes; dynamic programming reuses smaller answers
- Dijkstra settles the smallest number first
- Build a route fast, improve it with 2-opt, judge the longest ride
- Say the rules; the solver searches; you judge fairness

# Activity: Search race

1. Warm-up (3 min): Name one search from each lesson.
2. Race (10 min): Solve a small problem two ways.
3. Compare (4 min): Fast or sure? Same answer?
4. Fair (3 min): Who would the answer be hard on?

**Keep the problems small (four or five items, places or tasks) so both methods finish in the time.**

# Let’s discuss

1. When is a quick rule good enough?
2. Which method surprised you most?
3. Who should decide what “fair” means in a search?

# Exit ticket

- The method I would trust for my own search + one fairness question I would ask.